

# Pre-design solution: Тестовый отказ системы

**Region:** Новосибирская область **Greenhouse type:** year-round **Crop:** tomato **Target yield:** 2000.0 t/year **Generated:** 2026-06-17 04:07

## 1. Site inputs and analysis

Target yield per plot area: 4000.0 kg/m²/year  $\triangle$  Target yield is very high — intensive technology required.

**Climate parameters (per Russian SP 131.13330):**

| Parameter                      | Value         |
|--------------------------------|---------------|
| Design winter temperature (t5) | -37.0 °C      |
| Design summer temperature      | 25.0 °C       |
| Heating degree-days            | 6076.0        |
| Heating period                 | 227 days      |
| Snow load                      | 2.4 kPa       |
| Wind load                      | 0.3 kPa       |
| Solar radiation (winter)       | 1.5 MJ/m²/day |

## 2. Design solution (variant failed\_v1)

**Rationale:** A year-round multi-span block greenhouse complex is proposed for tomato trellis culture in the Novosibirsk region, where severe winters demand a fully enclosed, energy-efficient structure. Glass covering is selected to maximise light transmittance ( $\tau \approx 0.88$ ), which is critical given the very low winter solar radiation in the region. The brief mandates at least two separate blocks; however, the plot is extremely small (25 × 20 m) and can physically accommodate only two narrow blocks side by side. To satisfy the SP 107 clause 4.4 requirement of a minimum 6 m gap between year-round greenhouses, Block A and Block B are arranged along the 25 m plot length with a 6 m service corridor between them, leaving each block with a width of 7 m — the only geometrically feasible arrangement on this plot. Foundation choice: clay soil with groundwater at 0.5 m constitutes a frost-heaving, high-moisture condition, therefore pile foundations with perimeter drainage and enhanced waterproofing are specified. It must be noted that the 500 m² plot physically cannot support a 2 000 t/year tomato yield regardless of technology — this constraint is flagged for the Client's attention and the yield target should be revised to match the actual growing area.

**Total growing area:** 252 m<sup>2</sup> **Complex footprint (with auxiliary):** 252.0 m<sup>2</sup>

## Greenhouse blocks

- **Block A** — 18.0 × 7.0 m (area 126 m<sup>2</sup>)
- Heights: eave 2.4 m, ridge 5.5 m
- Layout: block (multi-span), 1 spans of 7.0 m
- Roof: straight, slope 45.0%
- Covering: glass ( $\tau=0.88$ ), glass 4.0 mm - Opaque structures share: 15.0%
- **Block B** — 18.0 × 7.0 m (area 126 m<sup>2</sup>)
- Heights: eave 2.4 m, ridge 5.5 m
- Layout: block (multi-span), 1 spans of 7.0 m
- Roof: straight, slope 45.0%
- Covering: glass ( $\tau=0.88$ ), glass 4.0 mm - Opaque structures share: 15.0%

## Complex structural parameters

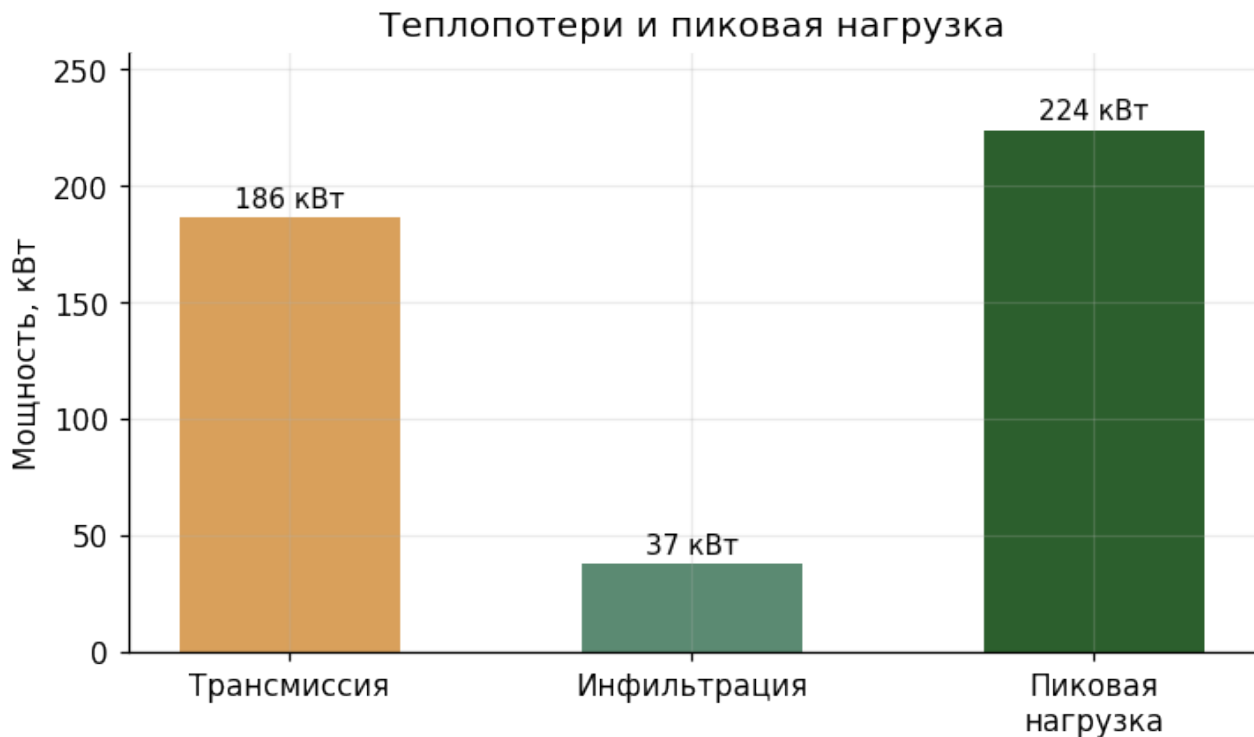
| Parameter                  | Value | SP Clause |
|----------------------------|-------|-----------|
| Plinth height              | 0.4 m | 5.8       |
| Foundation rise above soil | 0.4 m | 5.9       |
| Territory fence height     | 1.8 m | 4.16      |

## Auxiliary zones

- **Boiler house** — 24.0 m<sup>2</sup> (Autonomous heating plant (no gas/grid supply — biomass or diesel boiler))
  - **Water storage** — 12.0 m<sup>2</sup> (Rainwater/trucked-water collection tank and irrigation pump station)
  - **Service corridor** — 108.0 m<sup>2</sup> (6 m gap between blocks per SP 107 cl. 4.4, used as vehicle access and snow-clearing lane)
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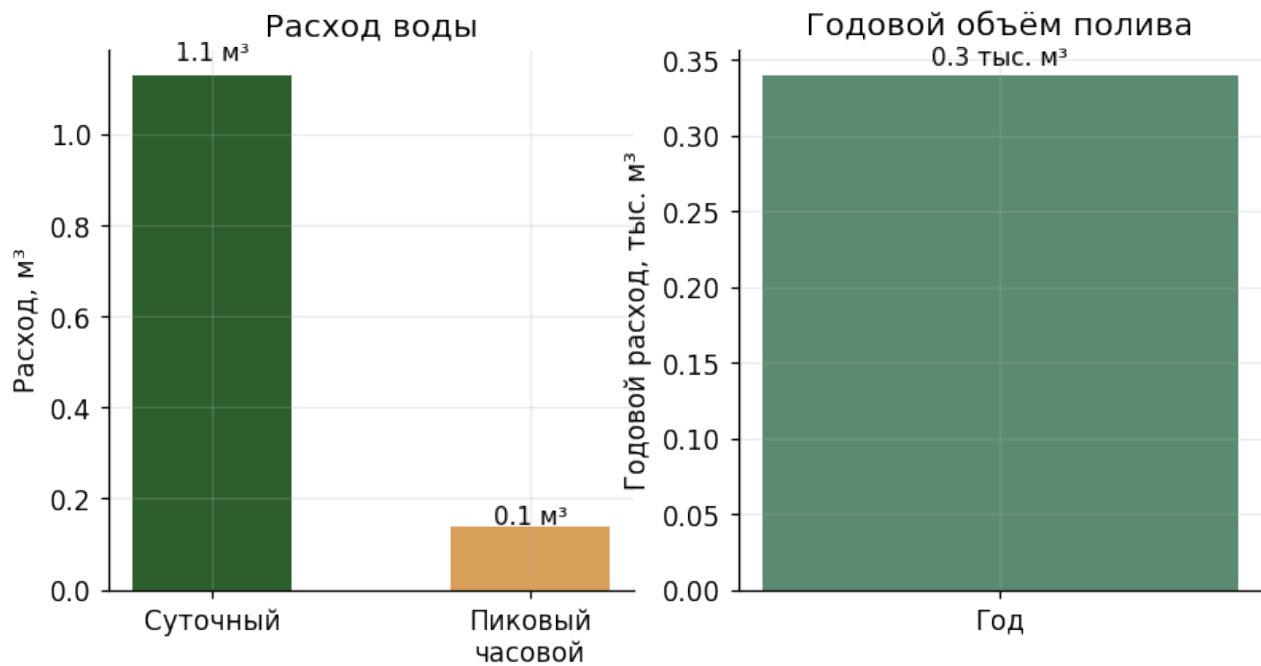
### 3. Engineering calculations

#### 3.1. Heat supply



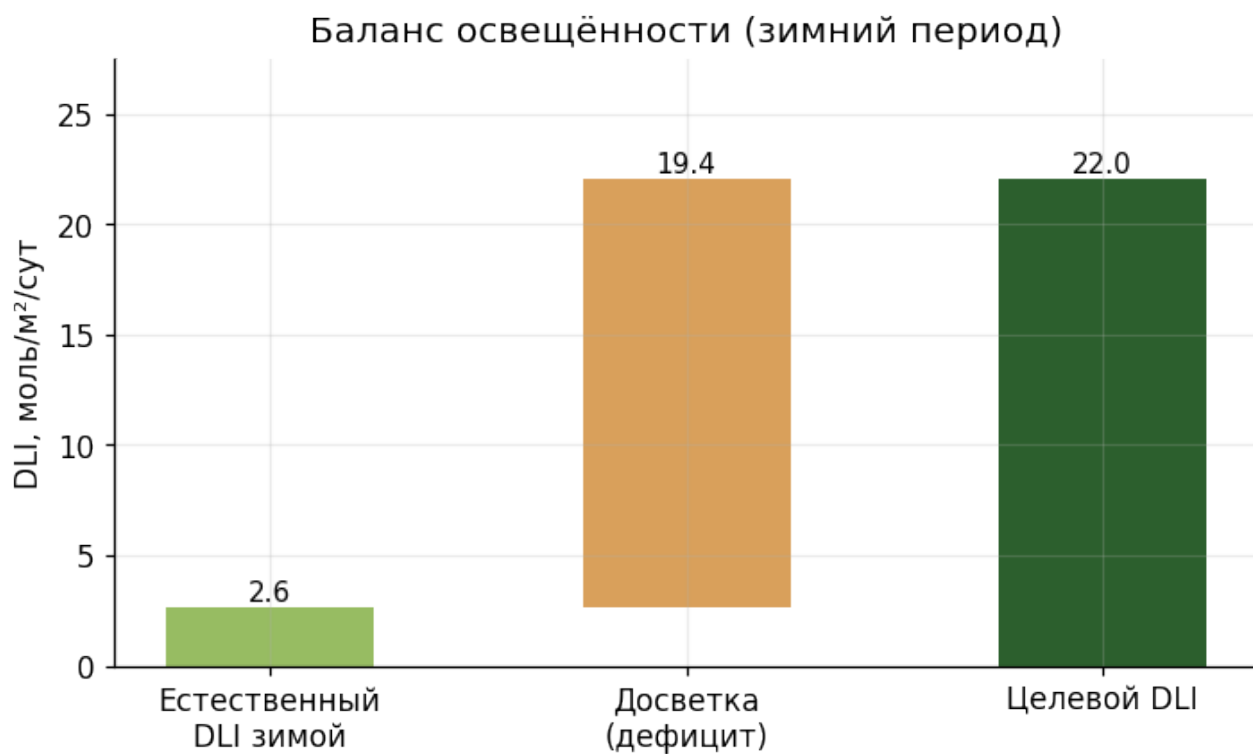
- Design temperature difference: **55.0 °C**
- Envelope area: **529.8 m<sup>2</sup>**
- Overall heat transfer coefficient U: **6.4 W/(m<sup>2</sup>·K)**
- Transmission losses: **186.5 kW**
- Infiltration losses: **37.3 kW**
- **Peak heating load: 223.8 kW**
- Annual heat demand: **593.3 MWh**
- Coolant temperature: **95.0 °C** (SP 7.9 ≤150)
- Lower-zone heat share: **45.0%** (SP 7.13 ≥40)

### 3.2. Water supply



- Daily demand: **1.13 m³/day**
- Peak hourly: **0.14 m³/h**
- Annual: **340 m³/year**
- Irrigation method: Drip (default)
- Reliability category: **II** (SP 6.14)
- Hose service radius: **40.0 m** (SP 6.8 ≤45)

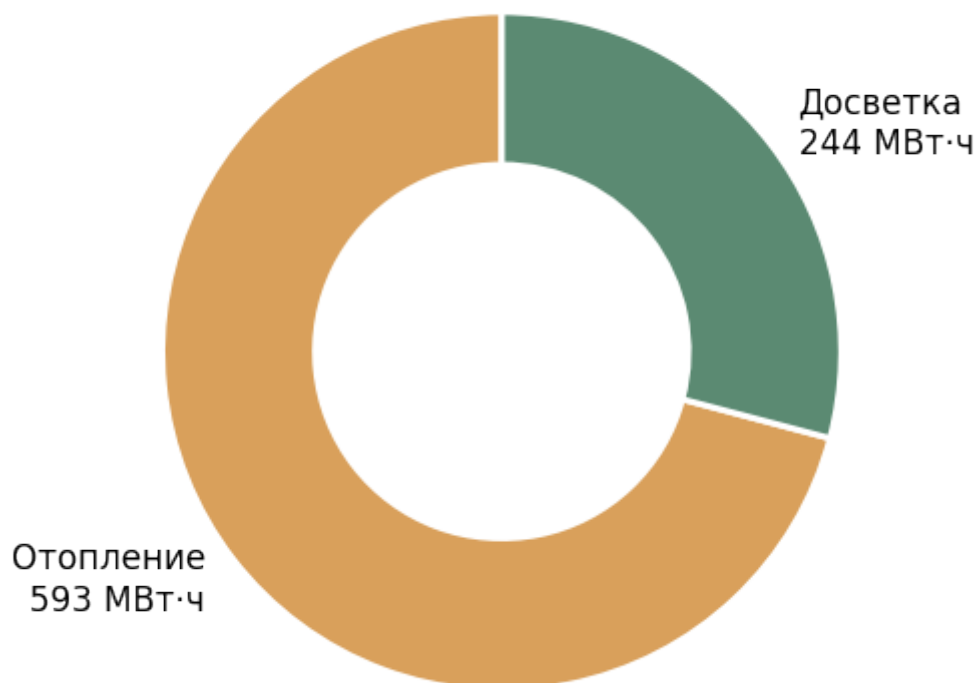
### 3.3. Lighting and supplemental



- Target DLI: **22.0 mol/m²/day**
- Natural winter DLI: **2.6 mol/m²/day**
- **Supplemental lighting required:** installed 266.2 W/m², consumption 243643 kWh/year
- Aisle floor illuminance: 8.0 lx (SP 8.3 ≤10)

### 3.4. Annual energy consumption

Годовое энергопотребление: 837 МВт·ч



### 3.5. Ventilation

- Summer target air changes per hour: **60.0 h<sup>-1</sup>**
- Vent opening share of envelope: **20.0%** (SP 7.18  $\geq 20$ ,  $\geq 10$  north of 60°N)
- Vent opening share of floor: 42.0%
- Forced ventilation: **not required**

### 3.6. Structural loads

- Snow load on roof: **974.0 kN** ( $\gamma=1.4$ )
- Wind load on walls: **58.0 kN** ( $q_{10}=1.0$ ,  $q_2=0.6$ )
- Trellis load: 150.0 N/m<sup>2</sup> ( $\gamma=1.3$ )

Overload factors and normative values per SP 107.13330 clause 5.14.

### 3.7. Geotechnical conditions and foundation

- Foundation soil: **clay**
- Groundwater depth: **0.5 m**
- Recommended foundation type: **pile**
- Minimum embedment depth: **3.0 m**

- Perimeter drainage: **required**
- Waterproofing grade: **enhanced**

Clay with high groundwater table — frost-heaving soil. Pile foundation embedded below the freezing depth; enhanced waterproofing and mandatory perimeter drainage required.

Preliminary recommendations. A proper engineering-geological survey per SP 47.13330 is required for working documentation.

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## 4. SP 107.13330 compliance check

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Rules checked: **36**

### WARNING — ENG.2-aux-share

Подсобные зоны должны занимать 5–30% площади комплекса. Меньше — нехватка для обслуживания; больше — неэффективное использование участка.

- Actual: 57.14285714285714
- Required: [5.0, 30.0]
- **Source** (SP 107.13330, original Russian text): Инженерная проверка (не из СП): обоснованная доля подсобок

### ERROR — ENG.5-yield-feasibility

Площадь выращивания × норма культуры должна покрывать минимум 90% целевой урожайности из ТЗ. Иначе участок физически не вмещает заявленный объём продукции.

- Actual: вмещает ~13 т/год (tomato @ 50.0 кг/м<sup>2</sup> × 252 м<sup>2</sup>)
- Required: >= 1800 т/год (90% от ТЗ 2000.0 т)
- **Source** (SP 107.13330, original Russian text): Инженерная проверка (не из СП): достижимость целевой урожайности

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Generated by agro-greenhouse-designer. Outputs are pre-design estimates and require validation by a qualified engineer before detailed documentation.